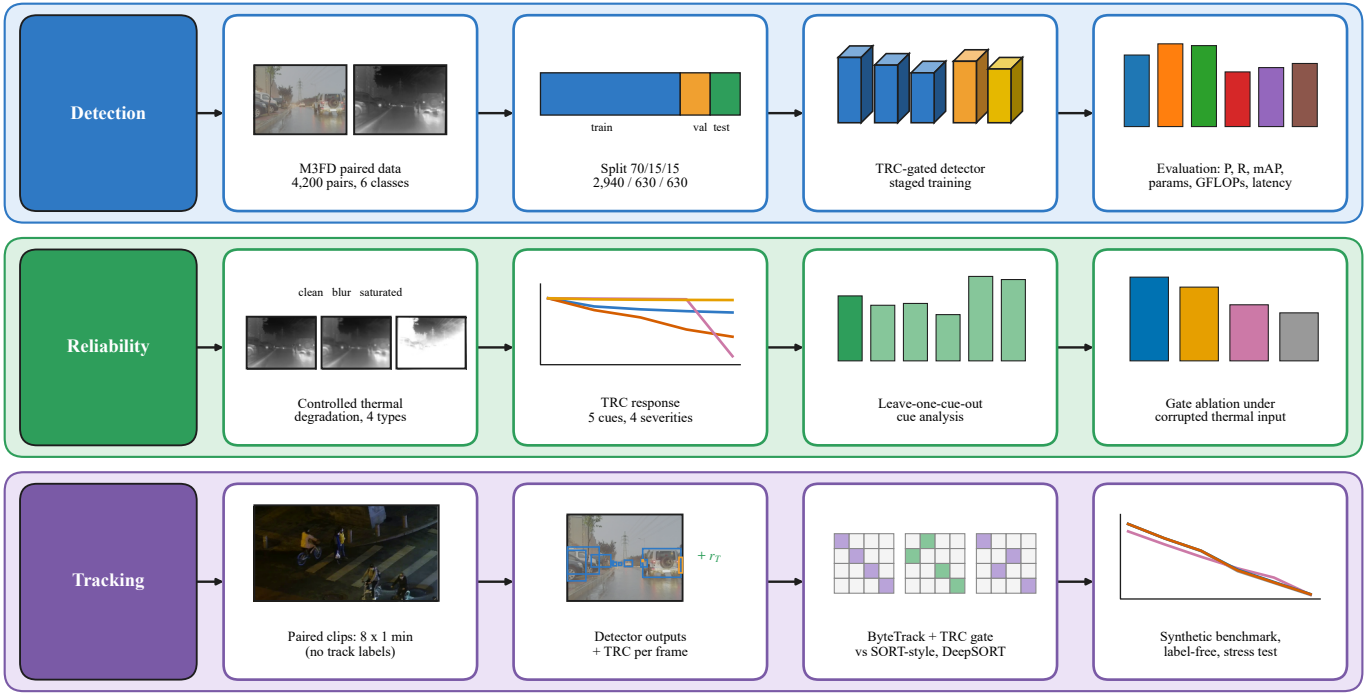

Detection
M3FD paired data
4,200 pairs, 6 classes
train
val test
Split 70/15/15
2,940 / 630 / 630
TRC-gated detector
staged training
Evaluation: P, R, mAP,
params, GFLOPs, latency
Reliability
clean blur saturated
Controlled thermal
degradation, 4 types
TRC response
5 cues, 4 severities
Leave-one-cue-out
cue analysis
Gate ablation under
corrupted thermal input
Tracking
Paired clips: 8 x 1 min
(no track labels)
+ r_T
Detector outputs
+ TRC per frame
ByteTrack + TRC gate
vs SORT-style, DeepSORT
Synthetic benchmark,
label-free, stress test